FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 1.2: CHEMICALS OF LIFE	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION: CHEMICALS OF LIFE
Biology Grade 10 | Sub-strand 1.2 | Demonstrating Complete Understanding

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Student Name: ___________________________________
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WHAT IS THIS DOCUMENT FOR?

You have now completed all 6 lessons of the *Chemicals of Life* unit. This Final Explanation is your chance to bring everything together and show that you truly understand — not just remember — the science you have investigated.

Think back to the two children in Kisumu. One child eats ugali every day and develops a swollen belly, thin limbs, discoloured hair, and extreme fatigue. The other child eats githeri — maize and beans together — and remains healthy and energetic. Both families have the same *amount* of food. Yet their bodies look completely different. Your job in this document is to fully explain why.

A strong Final Explanation does not just list facts. It uses evidence from your experiments and activities, connects ideas across all 6 lessons, and reasons clearly from what food is made of all the way to what happens inside a living body when a chemical is missing.

WHAT TO USE WHEN WRITING:

- Your Summary Table (completed across all 6 lessons)
- Your initial model sketch and your revised final model
- Notes, observations, and data from every lesson
- Food test results (specific colour changes and foods tested)
- Any readings, videos, or discussions from class

HOW YOU WILL BE GRADED:

Your Final Explanation will be marked out of **20 points** using the Final Explanation Rubric. The four criteria are:

1. Understanding of the Chemicals of Life (5 pts)
2. Use of Evidence from Experiments and Activities (5 pts)
3. Explaining the Phenomenon — the Two Children in Kisumu (5 pts)
4. Scientific Reasoning and Communication (5 pts)

LENGTH:

Aim for at least **4–5 pages handwritten** or **3–4 pages typed**. Every section below should be answered thoroughly. Short, vague answers will not score well. Specific evidence, named chemicals, named experiments, and clear reasoning will.

BEFORE YOU BEGIN — THE ANCHORING PHENOMENON

Read this carefully and keep it in mind as you write every section:

A Standard 5 pupil from Kisumu eats ugali every day — the same quantity of food as a neighbour who eats githeri. Yet the ugali-only child develops kwashiorkor: a swollen belly (caused by fluid retention from lack of plasma proteins), thin wasting limbs, discoloured reddish hair, and severe fatigue. The githeri child, whose meal contains both maize AND beans, remains healthy. How can the same quantity of food produce such different bodies?

Your explanation across all five parts below should, together, answer the driving question:

"Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?"

PART 1: What Are the Chemicals of Life — and What Is Food Really Made Of?

In this section, explain what biomolecules are, describe the main classes found in food, state what elements make up each class, and explain what each one does inside a living body. Use specific Kenyan food examples for each biomolecule. Make sure you connect this directly to the two children in Kisumu: which biomolecule is missing from a ugali-only diet, and why does that matter? **Your explanation should cover:** **– What the term 'biomolecule' means and why these chemicals are called 'chemicals of life'** **– Carbohydrates — elements, structure (monosaccharides → polysaccharides), functions (energy), and Kenyan food sources (ugali/maize, githeri, sweet potato, rice)** **– Proteins — elements (including nitrogen and sulfur), structure (amino acids → polypeptides), functions (growth, repair, enzymes, transport, immune response), and Kenyan food sources (beans, lentils, nyama choma, fish from Lake Victoria, eggs, milk)** **– Lipids — elements, structure (glycerol + fatty acids), functions (long-term energy store, insulation, cell membranes, fat-soluble vitamins), and Kenyan food sources (avocado, cooking**	Biomolecules are the chemical compounds produced by and found inside living organisms. They are called the 'chemicals of life' because every living process — growing, moving, digesting, fighting disease — depends on them. The four main classes of biomolecules found in food are carbohydrates, proteins, lipids, and nucleic acids, alongside vitamins, minerals, and water. Carbohydrates are made of carbon, hydrogen, and oxygen (C, H, O), always in an approximate ratio of 1:2:1. Simple carbohydrates are called monosaccharides (like glucose and fructose). When many glucose units join together, they form polysaccharides like starch. Ugali is made from maize flour, which is almost entirely starch. Carbohydrates are the body's primary and fastest source of energy — when you eat ugali, your digestive system breaks the starch down into glucose, which your cells use for cellular respiration. Proteins are made of carbon, hydrogen, oxygen, nitrogen, and sulfur (C, H, O, N, S). Their building blocks are amino acids, which link together in long chains called polypeptides that fold into specific 3D shapes. Proteins have many roles in the body: they build and repair muscle tissue, form enzymes that control chemical reactions, make antibodies that fight infection, and form haemoglobin that carries oxygen in the blood. Good Kenyan sources of protein include beans (as in githeri), fish from Lake Victoria, eggs, milk, lentils (ndengu), and nyama choma. Lipids (fats and oils) are also made of C, H, and O, but with far less oxygen than carbohydrates. They are built from one glycerol molecule joined to three fatty acid chains. Lipids store energy long-term, insulate the body, form the membranes of every cell, and carry fat-soluble vitamins (A, D, E, K). Kenyan sources include avocado, groundnuts, coconut, and cooking oil. Vitamins and minerals are needed in small amounts but are essential for regulating body processes. For example, Vitamin C (from oranges and mangoes) helps build connective tissue; iron (from red meat and leafy greens like sukuma wiki) is needed for haemoglobin; calcium (from milk and dagaa fish) builds strong bones. Water makes up about 70% of the human body. It acts as a solvent for all biochemical reactions, a transport medium for nutrients and wastes in the blood, a reactant in digestion (hydrolysis reactions break bonds using water molecules), and a temperature regulator through sweating. Now connect this to Kisumu: ugali supplies carbohydrates — energy — in abundance. But it contains almost no protein and no significant amounts of vitamins or minerals. The child eating only ugali gets enough calories to feel 'full', but their body is starved of amino acids. Without amino acids, the body cannot build plasma proteins (like albumin) that maintain fluid balance, cannot repair muscle tissue, cannot produce enzymes and antibodies. Over time, this protein deficiency produces kwashiorkor. The githeri child's diet includes beans, which supply protein (amino acids), making githeri a nutritionally complete meal. The difference between the two children is not the quantity of food — it is the chemical composition of that food.

oil, groundnuts, coconut) – **Vitamins and Minerals — examples (Vitamin C from citrus/mango, iron from meat/leafy greens, calcium from milk/bones), their roles as regulators and co-factors** – **Water — its roles as a solvent, transport medium, temperature regulator, and reactant in digestion** – **The critical insight: ugali is almost entirely starch (carbohydrate). It provides energy but contains very little protein. Githeri adds beans, which are rich in protein. This single chemical difference explains the two children's different health outcomes.**	

PART 2: Food Tests — How Can We Identify Biomolecules in Food?	
In this section, describe the food tests you conducted in the laboratory. For each test, state the reagent used, the colour change that shows a positive result, and which Kenyan foods you tested that gave positive results. Then explain what these results tell us about the chemical composition of foods like ugali and githeri. **Your explanation should cover:** – **The Benedict's Test for reducing sugars — reagent, original colour, positive result colour, why it works, which foods tested positive** – **The Iodine Test for starch — reagent, original colour,**	During Lesson 3, we conducted four chemical food tests to identify which biomolecules are present in different foods. These tests gave us real laboratory evidence about the chemical composition of the foods we eat every day in Kenya. The Benedict's Test detects reducing sugars (like glucose and maltose). We added Benedict's reagent (which is blue) to a food sample and heated it in a hot water bath. A positive result is a colour change from blue → green → yellow → orange → brick-red, depending on how much sugar is present. A stronger orange-red colour means more reducing sugar. We found that ripe banana and boiled githeri broth gave positive results, while raw maize flour gave only a weak green result (since it contains mostly starch, not free glucose). The Iodine Test detects starch. We added drops of iodine solution (which is brown-yellow) to food samples. A positive result is a dramatic colour change to blue-black. Ugali, maize flour, and bread all gave strong blue-black results, confirming they are rich in starch. Boiled beans gave no colour change, showing they contain very little starch. The Biuret Test detects proteins. We added sodium hydroxide (NaOH) and then copper sulphate solution ($CuSO_4$) to food samples. The reagent is blue. A positive result is a colour change to purple or violet. Boiled bean extract, egg white, and milk all turned purple-violet, confirming they are rich in protein. Ugali solution gave no colour change — confirming it contains very little protein. This is the critical laboratory evidence that supports our explanation of the Kisumu children: ugali fails the Biuret test; githeri (with beans) passes it.

positive result colour, which foods tested positive (ugali, maize flour, bread) **– The Biuret Test for proteins — reagent, original colour, positive result colour, which foods tested positive (beans, egg white, milk)** **– The Emulsion Test (or grease-spot test) for lipids — procedure, positive result, which foods tested positive** **– Interpreting your results: what do the food tests reveal about why githeri is more nutritious than ugali alone?** **– Limitations of food tests: what can the tests NOT tell you?**	The Emulsion Test detects lipids. We dissolved food samples in ethanol, then poured the mixture into water. If lipids are present, a cloudy white emulsion forms. Avocado and groundnut samples gave a cloudy emulsion. Ugali and bean extracts gave clear solutions, showing they contain little fat. These results are directly relevant to the phenomenon. They show us scientifically, using colour-change evidence, that ugali is composed almost entirely of starch (positive iodine, negative Biuret), while beans contain protein (positive Biuret, negative iodine). Githeri combines both. The food tests make visible what we cannot see with our eyes: the chemical difference between two meals that look similarly filling. However, food tests have limitations. They tell us which biomolecule class is present, but not the exact amount or the quality (e.g. whether all essential amino acids are present). They also cannot detect vitamins or minerals, which are equally important for health.

PART 3: Digestion and Enzymes — How Does the Body Break Food Down and Use It?	
In this section, explain what enzymes are, how they work, and which enzymes are responsible for breaking down each class of biomolecule during digestion. Use your catalase experiment as evidence for how enzymes function. Then explain how the products of digestion are absorbed into the blood, connecting this to what the body does with those nutrients. **Your explanation should cover:** **– What enzymes are (biological catalysts, protein nature, specificity)** **– The lock-and-key model of enzyme action (active site, substrate, enzyme-**	Enzymes are biological catalysts — they are proteins that speed up chemical reactions in living organisms without being used up themselves. Every enzyme has a specific 3D shape at its active site, which means it can only bind to one type of molecule (its substrate). This is described by the lock-and-key model: just as a specific key fits only one lock, a specific enzyme fits only one substrate. When the substrate enters the active site, an enzyme-substrate complex forms. The enzyme helps break or form chemical bonds, producing products. The enzyme is then released unchanged and ready to catalyse the next reaction. For example, salivary amylase (found in your saliva) binds to starch molecules and breaks them into maltose (a shorter sugar). Amylase works on starch — it would not work on protein because the shape of the active site does not match. The main digestive enzymes are: (1) Amylase — breaks starch (from ugali/maize) into maltose and then glucose. Found in saliva and the small intestine. (2) Protease (including pepsin in the stomach) — breaks proteins (from beans, meat, fish) into amino acids. (3) Lipase — breaks lipids into glycerol and fatty acids. Found in the small intestine, helped by bile from the liver. In our Lesson 4 catalase experiment, we placed a piece of fresh liver (or potato) into hydrogen peroxide (H_2O_2). We observed vigorous bubbling — the release of oxygen gas. This demonstrated that the enzyme catalase in the liver cells was breaking down hydrogen peroxide (a toxic waste product of metabolism) into water and oxygen very rapidly. The experiment showed us that enzymes are highly effective catalysts: the reaction barely happened without the liver, but was fast and visible with it. We also showed that boiled liver produced no bubbling — because heating had destroyed (denatured) the enzyme, confirming that enzymes are proteins sensitive to temperature.

substrate complex, products) – **Key digestive enzymes: salivary amylase (starch → maltose), protease/pepsin (proteins → amino acids), lipase (lipids → glycerol + fatty acids)** – **Evidence from the catalase experiment: what did you observe when liver/potato was added to hydrogen peroxide? What did this demonstrate about enzyme function?** – **How digested molecules are absorbed in the small intestine (villi and microvilli — increased surface area; glucose and amino acids absorbed into blood; fatty acids into lymph)** – **Why protein absorption matters for the Kisumu child: if no protein is eaten, no amino acids enter the blood, and the body cannot build or repair anything**	After enzymes break food down, the small molecules are absorbed into the blood through the walls of the small intestine. The inner lining of the small intestine is folded into millions of tiny finger-like projections called villi, each covered in even tinier microvilli. This enormously increases the surface area available for absorption. Glucose and amino acids are absorbed directly into the blood capillaries inside the villi and transported to cells around the body. Fatty acids and glycerol are absorbed into the lymphatic system. This brings us back to Kisumu. The child eating only ugali digests starch to glucose — so glucose enters the blood and provides energy. But because the ugali contains almost no protein, almost no amino acids enter the blood. The body has no raw materials to synthesise plasma proteins, repair muscle fibres, produce enzymes, or build antibodies. Over weeks and months, this chemical absence causes visible, devastating consequences — kwashiorkor.

PART 4: Factors Affecting Enzyme Activity — Why Does It Matter How Enzymes Work?	
In this section, explain the factors that affect how fast enzymes work — particularly temperature and pH. Use your experimental results from Lesson 5 as evidence. Then connect enzyme activity to nutrition and health: what happens to digestion when conditions are not optimal? **Your explanation should cover:** – **How temperature affects enzyme activity: increasing**	Enzymes do not work equally well under all conditions. Two of the most important factors that affect enzyme activity are temperature and pH. Temperature and Enzyme Activity: As temperature increases from cold toward the enzyme's optimum temperature, enzyme activity increases. This is because more heat energy causes more collisions between enzyme molecules and substrate molecules. For human digestive enzymes, the optimum temperature is approximately 37°C — the normal temperature of the human body. This is not a coincidence; our bodies maintain 37°C precisely because it is the temperature at which our enzymes work most efficiently. Above the optimum temperature, enzyme activity drops sharply. This is because high heat causes the enzyme's protein structure to unfold — the active site loses its specific shape, and the substrate can no longer fit. This permanent change is called denaturation. A denatured enzyme cannot be 'fixed' by cooling it back down — its shape is permanently lost. In our Lesson 5 experiment, the enzyme worked fastest at around 37–40°C, produced fewer bubbles at 10°C (too cold — fewer collisions), and produced no bubbles at 80°C (denatured). When we drew a graph of rate versus temperature, the curve rose to a peak and then

rate up to the optimum (~37–40°C for human digestive enzymes), then denaturation above the optimum — shape of the rate-vs-temperature graph **– How pH affects enzyme activity: each enzyme has an optimum pH (e.g. pepsin works best at pH 2 in the stomach; salivary amylase works best at pH 7 in the mouth; trypsin works best at pH 8 in the small intestine)** **– What denaturation means and why it is irreversible** **– Results from your Lesson 5 experiment: describe what you observed at low, optimum, and high temperatures; describe what you observed at different pH levels** **– Why the human body maintains a temperature of ~37°C (connection to homeostasis)** **– Real-world connection: fever, food safety, and why cooking destroys enzymes in food**	dropped steeply — not symmetrically. This asymmetric bell shape is the key evidence that denaturation is happening, not just slowing. pH and Enzyme Activity: pH also affects enzyme shape. Each enzyme has an optimum pH at which its active site is in the correct charge state to bind its substrate. Pepsin (in the stomach) works best at pH 2 — the highly acidic environment created by stomach acid. Salivary amylase works best at pH 7 (neutral — the mouth). Trypsin (in the small intestine) works best at pH 8 (slightly alkaline). In our experiment, we tested catalase at different pH levels and found it worked fastest near neutral pH (~7), with activity falling at very acidic (pH 2) or very alkaline (pH 10) conditions. Again, extremes of pH denature the enzyme. These findings connect directly to health. The human body works hard to maintain 37°C internally (homeostasis) because enzyme efficiency depends on it. A fever above 40°C starts to denature enzymes — this is why very high fevers are dangerous. Knowing that heat denatures enzymes also explains why cooking food destroys the enzymes naturally present in raw food, and why food stored in a cold refrigerator stays fresh longer (low temperature slows microbial enzyme activity). For the children in Kisumu, enzyme function is directly relevant: even if a child does eat some protein, poor health (including infections and fever) can reduce enzyme efficiency, slowing protein digestion and reducing amino acid absorption — making an already poor diet even less effective at nourishing the body.

PART 5: Bringing It All Together — Explaining the Phenomenon Completely	
This is the most important section. Use everything you have learned across all 6 lessons to write a complete, connected explanation of the driving question. Do not just list facts — explain the cause-and-effect chain from the chemical composition of food all the way to the visible health differences	The driving question asks: why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of? The answer lies entirely in chemistry. Food is not simply 'food' — it is a mixture of specific biomolecules, and each biomolecule plays a different, irreplaceable role in the body. The child eating only ugali is consuming mainly starch — a carbohydrate polysaccharide made of long chains of glucose. Starch is broken down by amylase enzymes (first in the mouth, then the small intestine) into glucose. Glucose is absorbed through the villi of the small intestine into the blood and used for cellular respiration — providing energy. So the ugali child has energy. But energy is only one thing the body needs.

between the two children. Reflect on how your understanding changed from Lesson 1 to now. Discuss real-world implications for nutrition, food policy, and community health in Kenya. **Your explanation should:** – **Directly answer the driving question in your own words** – **Trace the full chain: food composition → food tests evidence → digestion by enzymes → absorption → use by cells → health outcome** – **Explain specifically what kwashiorkor is and why it develops chemically** – **Reflect: what did you predict at the start of Lesson 1 versus what you now understand?** – **Apply this understanding to a real recommendation: what advice would you give to a family in Kisumu, Homa Bay, or Kericho to improve their diet using locally available, affordable foods?** – **Discuss one broader issue: why do some Kenyan children have protein deficiency even when food is available? (Consider food culture, food prices, crop choices, or education)**	The githeri child eats maize and beans together. The beans add proteins to the diet. Proteins are made of amino acids, which are broken down from dietary protein by protease enzymes (like pepsin in the stomach and trypsin in the small intestine). The amino acids are absorbed into the bloodstream and travel to every cell in the body. Cells use amino acids to build new proteins: muscle fibres for growth and movement, plasma proteins (like albumin) that regulate fluid balance, haemoglobin that carries oxygen, antibodies that fight disease, and digestive enzymes themselves. Our food tests provided laboratory evidence for this chemical difference. The ugali sample tested positive for starch (iodine test → blue-black) but negative for protein (Biuret test — no colour change). The bean sample tested positive for protein (Biuret test → purple-violet) but negative for starch. This colour-change evidence confirmed with our own eyes that these two foods have completely different chemical compositions, even though they look similarly filling on a plate. The child eating only ugali lacks amino acids. Over weeks, the body begins to break down its own muscle proteins to survive, causing thin, wasted limbs. Without plasma proteins like albumin to maintain osmotic balance, fluid leaks out of the bloodstream into body tissues, causing the characteristic swollen belly (oedema) of kwashiorkor. Without enough protein to make melanin and hair structural proteins, the hair loses colour and becomes brittle. Without enzymes and immune proteins being produced at normal rates, the child becomes fatigued and vulnerable to infection. Every visible symptom of kwashiorkor maps directly onto a missing protein function. At the start of this unit in Lesson 1, many of us thought that the amount of food was what mattered most. We predicted the children might have different appetites, or that one family was lying about how much food they ate. Now we understand that quantity is only part of the story — chemical composition is everything. Two meals can be identical in calories and portion size but completely different in their molecular makeup, and that molecular difference is what determines health. For families in Kisumu, Homa Bay, or Kericho, the practical recommendation is clear and achievable: mix protein sources with starchy staples. Beans, lentils (ndengu), cowpeas (kunde), and groundnuts are cheap, locally grown, and protein-rich. Eating githeri instead of plain ugali costs very little more but provides a complete mixture of biomolecules. Fish from Lake Victoria, eggs from local poultry, and dagaa (small dried fish) are also affordable protein sources available across the Nyanza region. However, understanding the chemistry also makes us ask a harder question: why do some children still suffer protein deficiency even when protein-rich foods are available? In many Kenyan communities, cultural habits, the high relative cost of beans during certain seasons, preference for processed maize products, and limited nutrition education all contribute. A scientific understanding of the chemicals of life must therefore be paired with community education and food policy — ensuring that families not only have access to diverse foods, but understand why dietary diversity is not a luxury, but a biological necessity.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
1. Understanding of the	Accurately describes all major	Accurately describes most biomolecules	Describes some biomolecules but with

Chemicals of Life (Biomolecules)	biomolecules (carbohydrates, proteins, lipids, vitamins, minerals, water) with correct elemental composition, structure, and functions. Uses specific, correct Kenyan food examples for each. Clearly distinguishes the chemical difference between ugali and githeri and explains why protein is irreplaceable. Uses correct scientific vocabulary throughout (e.g. amino acids, polysaccharide, hydrolysis, polypeptide).	with mostly correct compositions and functions. Includes Kenyan food examples for most classes. Explains the difference between ugali and githeri in terms of protein content. Minor errors or omissions (e.g. one biomolecule class missing or one function incorrect) do not undermine the overall explanation.	significant gaps, errors, or confusion (e.g. mixes up functions of proteins and carbohydrates, or cannot state which elements are in proteins). Kenyan food examples are missing or vague. The chemical difference between ugali and githeri is stated but not explained at the molecular level.
2. Use of Evidence from Experiments and Activities	Integrates specific, accurate evidence from at least four lessons. Food test results are cited with correct reagents, colour changes, and named foods tested. The catalase experiment is described with specific observations (bubbling, no bubbling when boiled). Temperature and pH graphs are referenced with specific data points or trends (e.g. 'optimum at ~37–40°C', 'no activity above 80°C'). Evidence is used to support claims, not just listed.	References evidence from at least three lessons. Most food test results are correctly stated (reagent and colour change). The catalase experiment is mentioned with a general observation. Temperature or pH experiment results are referenced but may lack specific values or graph descriptions. Evidence is mostly connected to claims rather than just listed.	References evidence from only one or two lessons, or makes factual errors about experimental results (e.g. wrong colour for Biuret test, wrong reagent for iodine test). Experiments are mentioned by name but observations are vague or missing. Evidence is listed without being connected to the explanation.
3. Explaining the Phenomenon — The Two Children in Kisumu	Provides a complete, cause-and-effect explanation of why the two children have different health outcomes, tracing the full chain: chemical composition of food → digestion by specific enzymes → absorption of specific molecules → use by cells → visible health outcomes. Correctly and specifically explains kwashiorkor in chemical terms (lack of amino acids → no plasma proteins → oedema; no structural proteins → muscle wasting; no pigment proteins → hair discolouration). Directly answers the driving question in the student's own words.	Explains why the two children differ with mostly correct reasoning. Connects protein deficiency to kwashiorkor symptoms, though the explanation of 2–3 specific symptoms may be incomplete or partially correct. The full chain from food to health outcome is present but one link may be unclear or missing. Driving question is addressed, if not always in precise scientific language.	Acknowledges that one child lacks protein and the other does not, but cannot explain why this causes the specific symptoms of kwashiorkor. The explanation focuses on symptoms without connecting them to missing molecular functions. The driving question is restated but not genuinely answered. Reasoning is mostly descriptive rather than cause-and-effect.
Scientific Reasoning, Reflection, and Real-World Application	Clearly reflects on how understanding changed from Lesson 1 predictions to the final explanation — showing genuine intellectual growth. Makes a specific, scientifically sound dietary recommendation for families in a named Kenyan region using affordable, locally	Reflects briefly on how understanding changed. Makes a general dietary recommendation that is scientifically reasonable and references Kenyan foods. Mentions a broader factor (e.g. food cost or culture) but does not develop it. Writing is mostly clear and organised,	Reflection is absent or superficial ('I learned a lot'). Dietary recommendation is too vague (e.g. 'eat more protein') or does not reference specific Kenyan foods. No broader social or economic factors are discussed. Writing is disorganised or difficult to follow, and

	available protein sources (e.g. beans, ndengu, dagaa, eggs). Discusses at least one broader social or economic factor that contributes to protein deficiency in Kenya. Writing is precise, well-organised, and uses scientific vocabulary accurately and consistently.	with occasional imprecise language but no major scientific errors.	scientific vocabulary is limited or frequently misused.